

PLANETARY NEBULA

# Hourglass Nebula



CATALOG NUMBER  
MyCn18

DISTANCE FROM SUN  
8,000 light-years

MAGNITUDE 11.8

MUSCA

The distinctive shape of the stunning Hourglass Nebula has caused much debate over its formation among astronomers. One suggestion is that as the aging, intermediate-mass star started to expand into a red giant, the escaping gas and dust accumulated first as a belt around the star's equator. As the volume of escaping gas continued to grow, the belt constricted the star's midsection, forcing the increasingly fast-moving gas into an hourglass shape. Other astronomers argue that the central star has a massive, heavy-element core that produces a strong magnetic field. In this scenario, the shape is a result of the ejected material being constrained by the magnetic field. Yet another suggestion is that the central star is in fact a binary and one of the pair is a white dwarf. A disk of dense material is produced around its middle by the gravitational interactions between the two components, which pinches in the "waist" of the expanding nebula. However, other features of the Hourglass Nebula have so far defied explanation. Astronomers have observed a second hourglass-shaped nebula within the larger one but, unusually, neither is positioned symmetrically around the central star. Two rings of material seen around the "eye" of the hourglass, perpendicular to one another, are the subjects of continuing studies.

EXPLORING SPACE

## NEBULA IN ACTION

The beautiful images of the Hourglass Nebula captured by the Hubble Space Telescope have revealed details within planetary nebulae that have revolutionized the study of these elusive but beautiful objects, especially in regard to the creation of nonspherical planetary nebulae. These fascinating nebulae are observed in many varied shapes, and an equally large number of hypotheses have been suggested to account for them. The life of a planetary nebula is a mere blink of an eye when compared to the lifetime of a star, but it is a very important stage. When a star is evolving off the main sequence, it loses huge quantities of its material and thus enriches the interstellar medium in elements heavier than helium, which can then be recycled to form other celestial objects.

### GAS SHELLS

This revealing picture of the Hourglass Nebula is a composite of three images taken in different wavelengths. The colorful gas rings are nitrogen (red), hydrogen (green), and oxygen (blue).